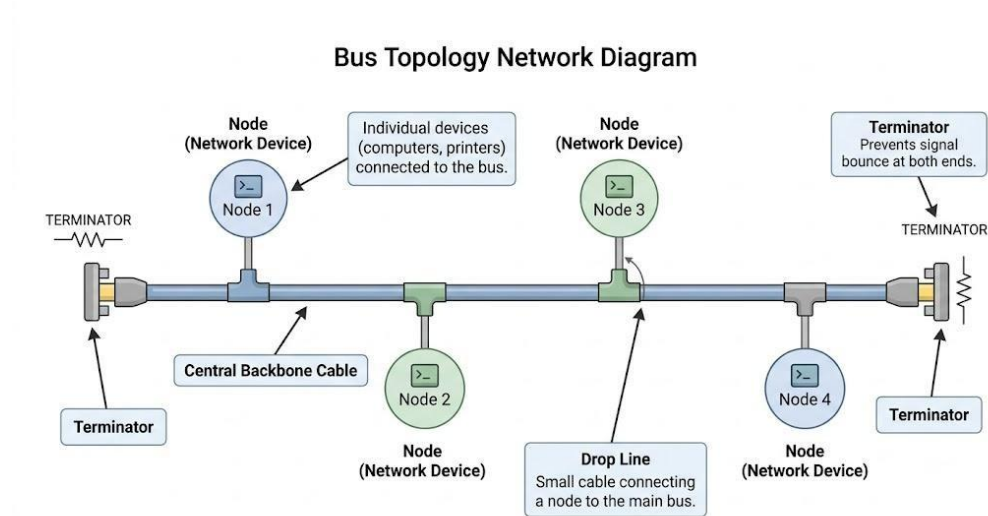


Day 2: Network Topologies Theory

Task 1 :- Draw a diagram of a bus topology and label all key components, then write one real-world example where this topology might be used.

ANS :-



Real-World Example : Automotive Controller Area Network (CAN Bus) :-

A major real-world application of a physical bus topology is the Controller Area Network (CAN bus) used in modern vehicles (cars, trucks, and buses).

- **How it works :**

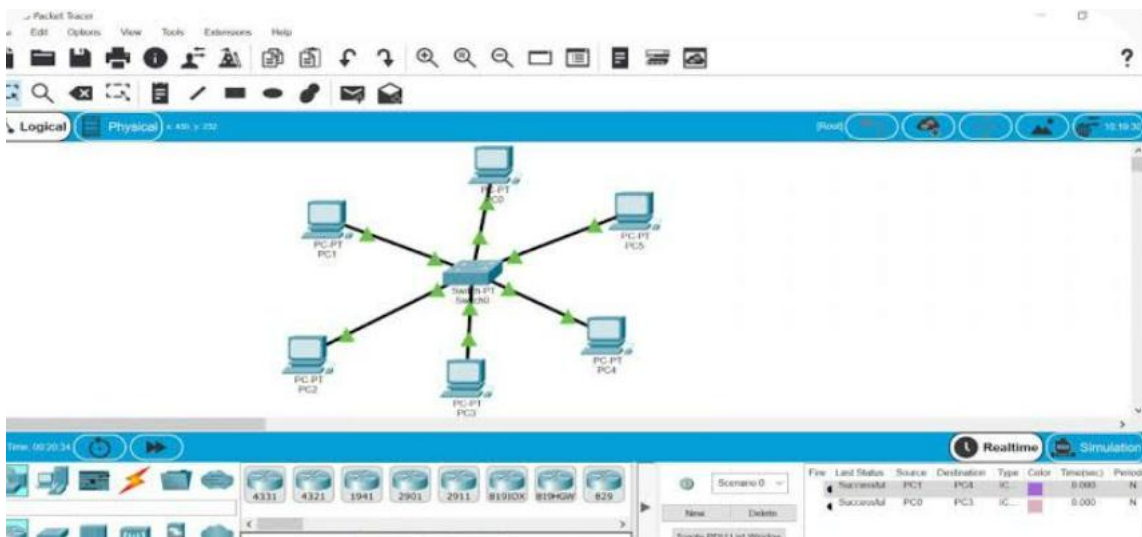
Instead of running miles of heavy wiring from every individual sensor to a central computer, all electronic control units (ECUs)—such as the engine control module, antilock braking system (ABS), airbag controllers, and power windows—are attached to a single shared pair of wires (the backbone cable) that runs throughout the vehicle.

- **Why bus topology is used :**

- **Drastically reduces weight and wiring complexity** : Saves space and cost compared to running separate wires to every component.
- **Reliable communication** : Allows sensors and computers to broadcast messages (like speed, temperature, or brake pressure) across the single shared bus so any control unit that needs that data can read it instantly.

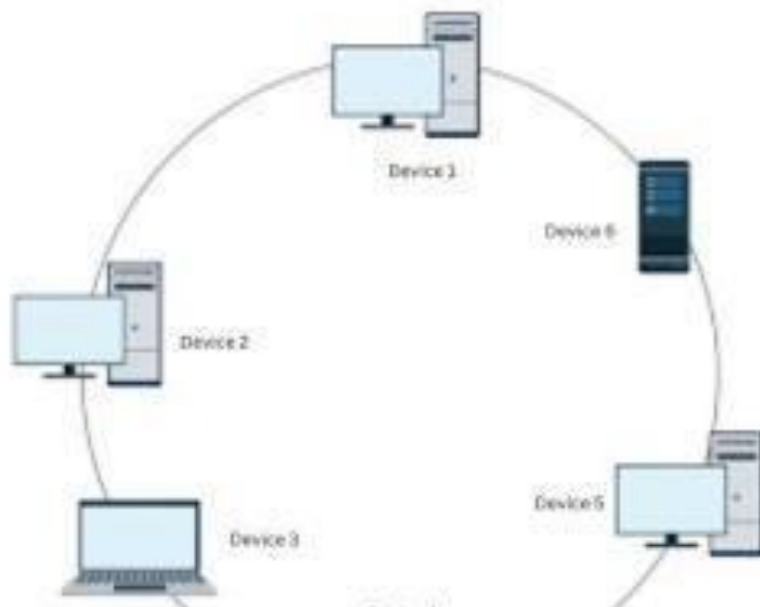
Task 2 :- In Cisco Packet Tracer, create a simple star topology with one switch and four PCs connected to it; name each PC after a popular app (e.g., WhatsApp, Zomato, Spotify, Flipkart) and take a screenshot of your topology.

ANS :-



Task 3 :- Draw a ring topology diagram and explain in 2-3 sentences how data travels between devices in this setup.

ANS :-



In a ring topology, each device is connected to exactly two neighboring devices, forming a continuous circular network loop. Data packets travel sequentially in one direction (clockwise or counterclockwise) from node to node around the ring. Every intermediate device reads the packet's destination address; if it is not the intended recipient, it retransmits the data to the next device until it reaches its target.

Task 4 :- Compare mesh and star topologies by listing 2 advantages and 2 disadvantages of each in a table format.

ANS :-

Topology	Advantages	Disadvantages
Star Topology	• Easy to fix : If one wire breaks, only that computer loses internet. • Simple to add devices : Just plug a new cable into the central switch.	• Single point of failure : If the central switch breaks, the whole network dies. • Depends on one device : You must buy a switch to connect everything.
Mesh Topology	• Super reliable : If one line cuts, data automatically takes another path. • High privacy : Messages travel directly between devices with no middleman eavesdropping	• Too many cables : Needs tons of wires, making it messy and very expensive. • Hard to set up : Connecting every device to every other device is extremely complicated.